

Last Time

$$\mathcal{H} := L^2(X_0, d\lambda_0, \mathbb{C}^4) = \{f: X_0 \rightarrow \mathbb{C}^4: f = (f_0, f_1, f_2, f_3) \text{ with } \int_{X_0} |f|^2 d\lambda_0 < \infty\}.$$

$$(\xi, \gamma) := - \int d\lambda_0(p) (\xi(p), \gamma(p)) \text{ for } \xi, \gamma \in \mathcal{H}.$$

↳ Inner Pr. is not positive-definite.

$$\mathcal{H}' := \{ \xi \in \mathcal{H} : (p, \xi(p)) = 0 \text{ for } \lambda_0\text{-a.e. } p \}.$$

⇒ Inner Pr. is positive-semidefinite.

$$\mathcal{H}_{\text{null}} := \{ \xi \in \mathcal{H}' : (\xi, \xi) = 0 \}.$$

$$\mathcal{H}_{\text{phys}} := \mathcal{H}' / \mathcal{H}_{\text{null}} : \text{Hilbert sp. under Minkowski posi-def.}$$

Hilbert sp. for a single photon.

$$(\alpha, A) \in \mathcal{P} = \mathbb{R}^{1,3} \rtimes SO^+(1,3).$$

For $\gamma \in \mathcal{H}$, define

$$(\mathcal{U}(\alpha, A)\gamma)(p) = e^{i(\alpha, p)} A\gamma(A^{-1}p)$$

sys. state

sys. state with new coord.

: representation of $\mathcal{P}$ in $\mathcal{H}$.

$$(a, A) \in \mathcal{P} = \mathbb{R}^{1,3} \times SO^\uparrow(1,3).$$

For $\psi \in \mathcal{H}$, define

$$(U(a, A)\psi)(p) = e^{i(a \cdot p)} A \psi(A^{-1}p).$$

system state

after changing the system state.

coordi. sys. by the action of (a, A) .

→ representation of $\mathcal{P}$ in $\mathcal{H}$.

Fact $U(a, A) : \mathcal{H}_{\text{phys}} \rightarrow \mathcal{H}_{\text{phys}}.$

λ_0 : Lorentz-invariant.

(4+) 1. $\mathcal{H}' \rightarrow \mathcal{H}'.$

Recall that $\mathcal{H}' := \{ \psi : (p, \psi(p)) = 0, \text{ for } \lambda_0, a. e. p \}$.

$$\int d\lambda_0(p) (p, U(a, A)\psi(p)) = \int d\lambda_0(p) (p, e^{i(a \cdot p)} A \psi(A^{-1}p))$$

$$= \int d\lambda_0(p) e^{i(a \cdot p)} (A^{-1}p, \psi(A^{-1}p)) \stackrel{(\circ)}{=} 0$$

⇒ $(p, U(a, A)\psi(p)) = 0, \text{ for } \lambda_0 a. e. p.$

∴ $U(a, A)\psi \in \mathcal{H}'.$

0 for $\lambda_0 a. e. p.$

2. $\mathcal{H}_{\text{phys}} \rightarrow \mathcal{H}_{\text{phys}}. (\text{i.e., } \mathcal{H}_{\text{null}} \rightarrow \mathcal{H}_{\text{null}})$

$$(U(a, A)\psi, U(a, A)\xi)$$

$$= - \int d\lambda_0(p) (e^{i(a \cdot p)} A \psi(A^{-1}p), e^{i(a \cdot p)} A \xi(A^{-1}p))$$

$$= - \int d\lambda_0(p) (A \psi(A^{-1}p), A \xi(A^{-1}p))$$

$$\stackrel{(\circ)}{=} - \int d\lambda_0(p) (\psi(A^{-1}p), \xi(A^{-1}p))$$

$$\stackrel{(\circ)}{=} - \int d\lambda_0(p) (\psi(p), \xi(p)) = (\psi, \xi). \quad \square$$

Inner product is Lorentz-invariant.

λ_0 is Lorentz-invariant.

The electromagnetic four-potential

$\mathcal{H} = L^2(X_0, d\lambda_0, \mathbb{C}^4)$ with Euclidean inner product.

$$(\xi, \gamma)_{\mathcal{H}} = \int d\lambda(p) \sum_{\mu=0}^3 \xi^\mu(p)^* \gamma^\mu(p).$$

Euclidean

$\Rightarrow \mathcal{H}$: Hilbert space under this inner product.

$\mathcal{B} :=$ Fock space of $\mathcal{H}$.

$\mathcal{H}$: basis $\{e_i\}$

= set of $\text{seq}_{n=0}^{\infty} (\gamma_0, \gamma_1, \dots)$, $\gamma_n \in \mathcal{H}_{\text{sym}}^{\otimes n}$
with $\sum_{n=0}^{\infty} \|\gamma_n\|^2 < \infty$, $\forall n$.

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basis: $\{e_{i_1} \otimes \dots \otimes e_{i_n}\}$

Define $B(\xi) = A(-\eta\xi)$, $B^\dagger(\xi) = A^\dagger(\xi)$

where $\eta = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix}$.

$\rightarrow B, B^\dagger$: operator-valued distributions on $\mathcal{H}$.
 $e_{i_1} \otimes \dots \otimes e_{i_n}$

$$[B(\xi), B(\gamma)] = [B^\dagger(\xi), B^\dagger(\gamma)] = 0$$

$$[B(\xi), B^\dagger(\gamma)] = [A(-\eta\xi), A^\dagger(\gamma)] = (-\eta\xi, \gamma)_{\mathcal{H}} \mathbb{1} = (\xi, \gamma)_{\mathcal{H}} \mathbb{1}.$$

$\rightarrow$ space of Schwartz $\mathcal{A}$ on $\mathbb{R}^4$ Fourier transform

For $f \in \mathcal{S}(\mathbb{R}^4)$, $\mu=0, 1, 2, 3$, define

$$A_\mu(f) = B(\hat{f} e_\mu) + B^\dagger(\hat{f} e_\mu), \in \mathcal{B}_0.$$

e_μ : μ th standard basis vector of $\mathbb{R}^{1,3}$.

$A = (A_0, A_1, A_2, A_3)$: quantized electromagnetic four-potential.

$$(*) \quad A(f) = \sum_{k=1}^{\infty} \alpha_k^* a_k$$

$$A^\dagger(f) = \sum_{k=1}^{\infty} \alpha_k a_k^\dagger$$

$$f = \sum_{k=1}^{\infty} \alpha_k e_k$$

$$a_k^\dagger |n_1, n_2, \dots\rangle = \sqrt{n_k+1} |n_1, n_2, \dots, n_k+1, n_{k+1}, \dots\rangle$$

where $a_k |n_1, n_2, \dots\rangle = \begin{cases} \sqrt{n_k} |n_1, n_2, \dots, n_k-1, n_{k+1}, \dots\rangle, & n_k \geq 1 \\ 0, & n_k = 0. \end{cases}$

Hilbert space $\mathcal{H}$, $\mathcal{B} := \{(\psi_0, \psi_1, \dots) : \psi_n \in \mathcal{H}_{\text{sym}}^{\otimes n} \text{ with } \sum_{n=0}^{\infty} \|\psi_n\|^2 < \infty\}$.

$$(\psi, \phi) = \sum_{n=0}^{\infty} (\psi_n, \phi_n).$$

$\Rightarrow \mathcal{B} : \text{Hilbert space} \rightarrow \text{Bosonic Fock sp. of } \mathcal{H}.$

24. The Electro Magnetic Field.

$\mathcal{B}$: bosonic Fock space of $\mathcal{H}$.

$\mathcal{B}'$: " of $\mathcal{H}'$.

$\Rightarrow \mathcal{B}'$: closed subspace of $\mathcal{B}$.

Define

$$(\xi_1 \otimes \dots \otimes \xi_n, \gamma_1 \otimes \dots \otimes \gamma_n) := \prod_{i=1}^n (\xi_i, \gamma_i).$$

$$\mathcal{B}_{null} := \{ \gamma \in \mathcal{B}' : \|\gamma\| = 0 \}.$$

$$\mathcal{B}_{phys} := \mathcal{B}' / \mathcal{B}_{null} \stackrel{\cong}{=} \text{Bosonic Fock sp of } \mathcal{H}_{phys}.$$

For $f \in \mathcal{F}(\mathbb{R}^4)$, $A_\mu(f)$ acts on $\mathcal{B}$, $\mu=0,1,2,3$.

But $A_\mu(f)$ not acts on $\mathcal{B}_{phys}$.

Because $A_\mu(f)$ not acts on $\mathcal{B}_{null}$.

Some linear combination of A_μ 's are well-defined on $\mathcal{B}_{phys}$.

ex1) $F_{\mu\nu} := \partial_\mu A_\nu - \partial_\nu A_\mu$. ($\partial_\mu = \frac{\partial}{\partial x^\mu}$)

$\Rightarrow$ For any $f \in \mathcal{F}(\mathbb{R}^4)$, $F_{\mu\nu}(f)$ is well defined on $\mathcal{B}_{phys}$.

$F = (F_{\mu\nu})_{0 \leq \mu, \nu \leq 3}$: "electromagnetic field."

$$\hookrightarrow \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{bmatrix} \quad \begin{array}{l} \text{Electro field } E = (E_x, E_y, E_z) \\ \text{Magnetic field } B = (B_x, B_y, B_z) \end{array}$$

ex2) $f_0, f_1, f_2, f_3 \in \mathcal{F}(\mathbb{R}^4)$ s.t

$$\partial_0 f_0 - \partial_1 f_1 - \partial_2 f_2 - \partial_3 f_3 = 0.$$

$\Rightarrow A_0(f_0) - A_1(f_1) - A_2(f_2) - A_3(f_3)$ acts on $\mathcal{B}_{\text{phys}}$.

> Classical (Maxwell) theory of electromagnetism.

In Maxwell's formulation, we have two vector fields.

$$\mathbf{E}: \mathbb{R}^{1,3} \rightarrow \mathbb{R}^3 \quad \text{"electric field"}$$

$$\mathbf{B}: \mathbb{R}^{1,3} \rightarrow \mathbb{R}^3 \quad \text{"magnetic field"}$$

Maxwell eqs. (2 (P, J) 쌍)

divergence

$$(1) \nabla \cdot \mathbf{E} = 4\pi \rho \quad \text{charge density.}$$

$$(2) \nabla \cdot \mathbf{B} = 0$$

curl

$$(3) \nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$(4) \nabla \times \mathbf{B} = 4\pi \mathbf{J} + \frac{\partial \mathbf{E}}{\partial t}$$

current density

$\rho, \mathbf{J}$ determine the evolution of electric and magnetic fields.

Assume no electron. i.e., $\rho \equiv 0, \mathbf{J} = 0$.

$\Rightarrow$ Maxwell's eqs in the vacuum.

$\rightarrow$ classical electromagnetic four-potential.

Take any smooth vector field

$$\mathbf{A}: \mathbb{R}^{1,3} \rightarrow \mathbb{R}^4, \quad \mathbf{A} = (A_0, A_1, A_2, A_3)$$

Define the "classical electromagnetic field" $F = (F_{\mu\nu})$ as $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. $0 \leq \mu, \nu \leq 3$.

Define two v.f. $\mathbf{E}$ & $\mathbf{B}$ as

$$\mathbf{E} = (E_x, E_y, E_z)$$

$$\mathbf{B} = (B_x, B_y, B_z)$$

$$F = \begin{bmatrix} 0 & E_x & E_y & E_z \\ -E_x & 0 & -B_z & B_y \\ -E_y & B_z & 0 & -B_x \\ -E_z & -B_y & B_x & 0 \end{bmatrix}$$

By the def. of F ,

We have an identity, called "Bianchi identity".

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0 \quad \text{for } \mu, \nu, \lambda.$$

And the v.f. B & E satisfy (2), (3) in Maxwell eq.

• Gauss's law for magnetism. (2)

$$\begin{aligned}\nabla \cdot B &= \partial_x B_x + \partial_y B_y + \partial_z B_z \\ &= -\partial_x F_{23} + \partial_y F_{13} - \partial_z F_{12} \\ &= \partial_x F_{32} + \partial_y F_{13} + \partial_z F_{21} \\ &\stackrel{\textcircled{=}}{=} \partial_1 F_{32} + \partial_2 F_{13} + \partial_3 F_{21} = 0.\end{aligned}$$

$\uparrow$
 $x=1, y=2, z=3$

• Maxwell-Faraday equation. (3)

$$\begin{aligned}\nabla \times E &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix} \\ &= (\partial_y E_z - \partial_z E_y, -\partial_x E_z + \partial_z E_x, \partial_x E_y - \partial_y E_x) \\ \partial_y E_z - \partial_z E_y &= \partial_y F_{tz} - \partial_z F_{ty} = \partial_y F_{tz} + \partial_z F_{yt} = \partial_t F_{yz} \\ &= \partial_t F_{23} = \partial_t (-B_x) \\ \therefore &= -(\partial_t B_x, \partial_t B_y, \partial_t B_z).\end{aligned}$$

Adding the condition, so called "equation of evolution"
 $\partial_0 F_{0\nu} - \partial_1 F_{1\nu} - \partial_2 F_{2\nu} - \partial_3 F_{3\nu} = 0 \quad \text{for } \nu.$

We can see B & E also satisfy (1), (4) in Maxwell equations in the vacuum.

- Gauss's Law.

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \partial_x E_x + \partial_y E_y + \partial_z E_z \\ &= \partial_x F_{tx} + \partial_y F_{ty} + \partial_z F_{tz} \\ &= -\partial_x F_{xt} - \partial_y F_{yt} - \partial_z F_{zt} = -\partial_t F_{tt} = 0\end{aligned}$$

- Ampère's Circuital Law.

$$\begin{aligned}\nabla \times \mathbf{B} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \partial_x & \partial_y & \partial_z \\ B_x & B_y & B_z \end{vmatrix} = (\partial_y B_z - \partial_z B_y, \partial_z B_x - \partial_x B_z, \partial_x B_y - \partial_y B_x) \\ \partial_y B_z - \partial_z B_y &= \partial_y (-F_{xz}) - \partial_z (-F_{xy}) = \partial_y F_{zx} + \partial_z F_{xy} \\ &= \partial_t F_{tx} - \partial_x F_{tx} \\ &= \partial_t E_x \\ &\rightarrow = (\partial_t E_x, \partial_t E_y, \partial_t E_z) .\end{aligned}$$

~~Fact~~ Any solution of Maxwell's equations in the vacuum is representable in the above manner.

But it's not unique.

The procedure for choosing a unique representative is called "gauge fixing".

One way to do this is to impose "Lorenz gauge condition",

$$\partial_0 A_0 - \partial_1 A_1 - \partial_2 A_2 - \partial_3 A_3 = 0.$$

$\therefore A = (A_\mu)_{0 \leq \mu \leq 3}$ defined by using Gupta-Bleuler.
satisfies the vacuum Maxwell equations
and the Lorenz gauge condition.

Einstein Notation (Summation Convention).

$$y = \sum_{i=1}^3 c_i x^i \quad \Rightarrow \quad y = c_i x^i.$$

For (a_0, a_1, a_2, a_3) ,

$$a^{(0)} := a_0, \quad a^{(\mu)} := -a_\mu, \quad \mu = 1, 2, 3.$$

$$\partial^\mu A_\mu := \sum_{\mu=0}^3 \partial^\mu A_\mu = \partial_0 A_0 - \partial_1 A_1 - \partial_2 A_2 - \partial_3 A_3.$$

ex) Lorenz gauge condition : $\partial^\mu A_\mu = 0$.
Minkowski inner product : $x^\mu y_\mu$.

가우스 법칙

$$\nabla \cdot D = \rho$$

Gauss's law

가우스 자기 법칙

$$\nabla \cdot B = 0$$

Gauss's law for magnetism

패러데이 전자기 유도 법칙

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

Maxwell-Faraday eq.

앙페르 - 맥스웰 회로 법칙

$$\nabla \times H = J + \frac{\partial D}{\partial t}$$

Ampère's circuital law.

E : 전기장, H : 자기장

D : 전기변위장

B : 자기장

electric field

magnetic field.

ρ : 자유전하밀도

J : 자유전류밀도

charge density

electric current density.

진공상의 전자기장.

$$\nabla \cdot E = \frac{\rho_v}{\epsilon_0}$$

전기장

volume charge density

$$\nabla \cdot B = 0$$

자기장

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \times B = \mu_0 \left(J_v + \epsilon_0 \frac{\partial E}{\partial t} \right)$$

volume current density.

C

$$S: \sup_{\alpha, \beta, \gamma} \|x^\beta \partial_\alpha^\gamma f\| < C$$

$$e^{-x^2}$$

$$\sigma \rightarrow S$$

$$S \subseteq \mathcal{H}.$$

$$\vec{e}_0 \subseteq S \subseteq L^2.$$

dense